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Matricule: **CT24PO16**

Topic : **A mobile-integrated deep learning system for rapid screening of respiratory diseases using cough sound analysis.**

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Table Summarising the Author and year, title, methodology, results and key limitations of the various articles below:

Author and Year	Title	Methodology	Results	Key Limitations
Ahmed et al. (2025)	Ensemble Deep Learning Models for Respiratory Disease Detection Using Cough Analysis	CNN-MLP-RNN fusion architecture; XGBoost integration; Mel Filter Bank coefficients; SpecAugment for data augmentation	99.94% accuracy ; 99.49% with XGBoost alone; Best performance achieved to date	Limited to 3 disease classes; High computational requirements; Lab conditions only
Barua et al. (2025)	Automated Lightweight Model for Asthma Detection Using Respiratory and Cough Sound Signals	Dual acoustic signals (cough + respiratory); SVM with SqueezeNet/EfficientNet; NCA feature selection	92% accuracy; >95% AUC; 89% sensitivity, 90% specificity	Limited to asthma/COPD only; Single-center validation; Small dataset (120 subjects)
Bousslama et al. (2025)	Toward AI-Driven Cough Sound Analysis for Respiratory Disease Diagnosis	Transfer learning with MFCC; CNN-SVM hybrid; Android app deployment; Real-time processing	99.55-100% validation accuracy; Functional mobile application	COVID-19 specific; Android platform only; Privacy concerns
Topaloglu et al. (2025)	Machine Learning-Driven Lung Sound Analysis for Asthma Diagnosis	MFCC-TQWT features; ReliefF selection; Quadratic SVM; Narrow Neural Network	99.86% accuracy ; >99% sensitivity/specificity; 14kB model size	Single disease focus; Limited to controlled asthma; Device-specific
Moon et al. (2025)	Machine Learning-Driven Strategies for Enhanced Pediatric Wheezing Detection	Spectral contrast features; t-SNE dimensioning; Kernel SVM classification	89.7% accuracy; 95.2% recall for wheeze detection	Pediatric-specific; Limited to wheeze only; Small sample (103 wheezes)

Fraihi et al. (2025)	Improving Deep Learning Respiratory Sound Analysis with Frequency Selection	CNN with temporal self-attention; Frequency band selection; Age-adaptive training	50% computational cost reduction; Improved convergence	ICBHI dataset only; Age-stratification needed; Complex implementation
Dietrich et al. (2025)	Evaluating Few-Shot Prompting for Lung Sound Classification Using GPT-4o	Multimodal LLM (GPT-4o); Mel-spectrogram s; Few-shot vs zero-shot prompting	36.3% accuracy (few-shot) vs 29.6% (zero-shot)	Very low accuracy vs specialized models; General-purpose model limitations
HeAR Team (2025)	Health Acoustic Representations - Self-Supervised Learning System	Masked autoencoders; 313 million audio clips training; Linear probes evaluation	State-of-the-art on 33 health acoustic tasks; Universal health audio embeddings	Requires massive data; Not disease-specific; Generalist approach
Koo et al. (2025)	Multimodal Respiratory Sound Classification with Counterfactual Debiasing	Counterfactual adversarial debiasing; Multimodal fusion; Out-of-distribution robustness	Improved generalization across clinical environments	Complex training; Requires metadata; Computational overhead
Liu et al. (2024)	Vision Transformers for Respiratory Disease Diagnosis	Self-supervised + supervised learning; Vision transformer architecture; Large-scale cough dataset	92.5% AUROC for COPD; Outperforms prior methods	High computational requirements; Limited to COPD; Need large datasets
Matsumura et al. (2024)	Real-time Healthcare Analysis Using Edge Computing	Multimodal sensor patch; Smartphone edge computing; Reservoir computing	>80% accuracy; Real-time processing achieved	Training on computer required; Only 3 subjects tested; Limited validation
Xu et al. (2024)	Listen2Cough: End-to-End Deep Learning Model	Deep CNN architecture; Passive audio sensing; Mobile	Enhanced lung health assessment; Good external	Battery consumption issues; Background

		optimization	validation	noise sensitivity; Privacy concerns
Kevser & Kaya (2024)	Cough Detection Using Acceleration Signals	Accelerometer-based; SqueezeNet with wavelets; Multiple DL architectures	92.21% accuracy, 95.59% precision; Low computational burden	Voluntary coughs only; Healthy subjects; No disease detection
Kapetanidis et al. (2024)	Systematic Review of Audio Analysis and AI	Review of 75 studies; ML/DL comparison; Trend analysis	Identified DL superiority; Comprehensive overview	Review paper only; No original experimentation
Hamdi et al. (2023)	Attention-based CNN-LSTM for COVID-19	Hybrid architecture; SpecAugment; Attention mechanisms; COUGHVID dataset	76.81% accuracy; 85.35% AUC on DiCOVA challenge	Class imbalance; COVID-19 specific only; Limited generalization
Askari Nasab et al. (2023)	Coronavirus Diagnosis Using Cough Sounds	Three-stage deep learning; Transformer + RNN; Mobile app deployment	78% recall; 30% false alarm rate; Real-time capability	High false positive rate; Limited to COVID-19
Melek (2023)	Edge Computing for Chronic Respiratory Diseases	Raspberry Pi/smartphone; Real-time classification; 86 subjects dataset	90% sensitivity, 93.55% specificity; 91.75% balanced accuracy	Small dataset; Limited disease types; Hardware dependent
Pahar et al. (2023)	NLP-Style TB Detection Using Cough Embeddings	Self-supervised learning; Transfer learning; ICBHI 2017 dataset	7.67% improvement over baseline; Better pattern capture	TB-specific only; Computational complexity; Limited diseases
Hsu et al. (2021)	RNN Variants for Breath Phase Detection	8 RNN variants tested; Open-access HF_Lung_V1 database; LSTM/GRU comparison	BiGRU best performance; Large open dataset created	Focus on breath phase only; Limited to detection not diagnosis
Perna & Tagarelli	Deep	Recurrent neural	Competitive	Older

(2019)	Auscultation with RNNs	networks; ICBHI dataset; Respiratory anomaly prediction	performance on benchmark	architecture; Limited disease coverage
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❖ New orientation for your topic

➤ Proposed main objective

Design and validate a **mobile-integrated deep learning system** that performs **rapid respiratory disease screening from cough audio**, with **on-device (or edge-assisted) inference**, and **explainability + privacy safeguards**, evaluated on realistic, noisy conditions and diverse devices.

Proposed specific objectives

1. **Dataset strategy:** assemble/curate cough recordings (healthy + multiple respiratory conditions) with device/environment metadata; define labeling rules and exclusion criteria.
2. **Robust cough event detection:** implement a lightweight detector optimized for phones (XAI-guided cropping / TFLite) to isolate cough segments reliably.
3. **Disease screening model:** train a multi-class classifier using transfer/self-supervised learning and compare CNN vs transformer families (including mobile-efficient backbones).
4. **Edge/mobile optimization:** quantization/pruning + latency/battery profiling; define acceptable trade-offs for real-time use.
5. **Explainability & safety:** include interpretable heatmaps/feature-attribution for cough segments; build guardrails for “uncertain” outputs.
6. **Validation plan:** test generalization across devices/noise; report AUROC/F1/sensitivity/specificity + calibration; include protocol-style evaluation plan.

Timeline (December → May)

December

- Finalize research questions + target diseases/classes
- Define inclusion/exclusion + annotation protocol
- Reproduce 2–3 baseline pipelines (classic CNN, CRNN, transformer-lite)

January

- Data acquisition/curation + cleaning + cough segmentation pipeline
- Train cough detector; optimize for mobile constraints (TFLite-ready)

February

- Train multi-class screening models (CNN vs transformer; transfer/self-supervised)
- Cross-device/noise augmentation and ablations

March

- Mobile app integration (end-to-end pipeline: record → detect → classify)
- On-device optimization: quantization/pruning + latency/battery tests

April

- Validation study runs (held-out device sets; robustness testing)
- Explainability module + reporting templates (tables/figures)

May

- Final evaluation + thesis write-up: results, limitations, and deployment guidance

- Prepare demo + documentation + publication-ready formatting